A PROPER SPACE FINITE LOCALLY FREE OVER A DENSE OPEN BECOMES FINITE LOCALLY FREE AFTER AN ADMISSIBLE BLOW-UP

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 8912-8924 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 8926-8934, the complete author proof (7 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. This is the input cited by name in the spaces resolution-property capstone L16104 (this wave); the lane sequence keeps dependencies on the shelf.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemmas flat-after-blowing-up (the flattening engine) and proper-flat-finite-over-dense-open (card L16101, on the shelf); Morphisms of Spaces, lemma finite-flat (finite flat finitely presented modules are finite locally free); admissible blow-up composition.

Source reference. Original lemma 087E: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8861–8869. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0B4J: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8625–8631. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0416: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8934–8934. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to B$ be a morphism of algebraic spaces over S. Let $U \subset B$ be an open subspace. Assume*

(1) *B is quasi-compact and quasi-separated,*
(2) *U is quasi-compact,*
(3) *$f : X \to B$ is proper, and*
(4) *$f^{-1}(U) \to U$ is finite locally free.*

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Then there exists a U-admissible blowup $B' \to B$ such that the strict transform X' of X is finite locally free over B'.

Proof. By Lemma 087E we may assume that $X \to B$ is flat and of finite presentation. After replacing B by a U-admissible blowup if necessary, we may assume that $U \subset B$ is scheme theoretically dense. Then f is finite by Lemma 0B4J. Hence f is finite locally free by Morphisms of Spaces, Lemma 0416. □